

Which generalized entropy? Discrimination among horizon-entropy dark-energy models with DESI DR2 and Planck, and the closure of the structure-growth channel

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Abstract

Horizon-entropy (holographic) dark energy is a large and active post-DESI program, but the literature is fragmented: each study fits a single entropy prescription and reports that it can reproduce the data, leaving open whether the data *discriminate* between prescriptions. Using a uniform pipeline applied to real DESI DR2 baryon-acoustic-oscillation data and Planck cosmic microwave background distance priors, we show that they do. Casting Barrow and Tsallis holographic dark energy as a single power-law family $\rho_{\text{DE}} \propto L^n$ and treating Rényi holographic dark energy as its genuinely distinct coupled generalization, we find that the *direction* of entropy deformation is decisive: steeper $\rho(L)$ (Tsallis $\delta_T < 1$) relieves the tension of standard holographic dark energy with the data, while flatter $\rho(L)$ (Barrow $\Delta > 0$) worsens it; no variant is preferred over Λ CDM. We further show, analytically, that the linear structure-growth rate carries no information about these models beyond the background expansion: for any event-horizon holographic dark energy, sub-horizon dark-energy perturbations are suppressed relative to matter as $1/k^3$ by phase-averaging of the gravitational potential along the future light cone, so $f\sigma_8$ cannot separate the models from one another or from Λ CDM beyond what background geometry already achieves. The proposed structure-growth confrontation of these models is therefore, for this class, structurally empty.

1 Introduction

The thermodynamic route to cosmology—in which the Friedmann equation is recovered as the first law of thermodynamics applied to the apparent horizon—has generated a broad family of dark-energy models built from generalized (non-additive) entropies. Replacing the Bekenstein–Hawking area law $S \propto A$ with a deformed law (Barrow, Tsallis, Rényi, Kaniadakis, Sharma–Mittal, ...) and identifying an infrared cutoff L yields a holographic dark-energy (HDE) density $\rho_{\text{DE}} \sim L^n$ or a generalization thereof [1, 2, 3, 4]. Post-DESI, this has become a crowded program.

The crowding, however, hides a gap. Individual studies typically fix one entropy prescription, fit it, and report that it can accommodate the data. What is missing is a *uniform* treatment that asks the comparative question: given the same data and the same pipeline, do the data prefer one entropy deformation over another, or are they all equivalent phenomenological fits? We answer this here.

A second, sharper motivation comes from the fitting literature itself. Confronting Kaniadakis holographic dark energy with DESI DR2, cosmic chronometers, and supernovae, Luciano et al. [8, 9] find a structural degeneracy between the holographic parameters and the matter density at the background level, and conclude explicitly that expansion data alone are insufficient to discriminate the entropy deformation, so that *perturbation observables*

are essential. This is precisely the call our second result addresses. We show that, for event-horizon holographic dark energy, the structure-growth channel carries no information beyond the background—so the perturbation observables the field is turning to are, for this class, empty of discriminating power.

This paper is a companion to Ref. [13], which developed a detectability and exclusion framework for localized features in the dark-energy expansion history and, as a worked example on a physical theory, constrained the Barrow deformation of standard HDE. We inherit that paper’s validated CMB distance-prior likelihood and its framing of what expansion-history geometry can and cannot resolve, and we extend the analysis from a single model to the model *family*, adding two results. First, a background-level discrimination: the data pick a side (Sec. 3). Second, and independently, a negative result at the perturbation level: the structure-growth channel is closed for the whole family (Sec. 4). Together these complete the picture—the models are fully testable with present data at the background level, and growth data add nothing.

Throughout we adopt a spatially flat universe with $\Omega_{m0} = 0.31$ unless stated otherwise, and we use natural units in which $3/(8\pi G) = H_0 = 1$ where convenient.

2 A unified framework for horizon-entropy dark energy

2.1 Barrow and Tsallis as one power law

Barrow entropy, $S \propto A^{1+\Delta/2}$ with $\Delta \in [0, 1]$ a quantum-gravitational deformation parameter, and Tsallis entropy, $S \propto A^{\delta_T}$, both lead—through the holographic construction with a future-event-horizon cutoff—to a power-law dark-energy density

$$\rho_{\text{DE}} \propto L^n, \quad L = R_h = a \int_t^\infty \frac{dt'}{a(t')}, \quad (1)$$

with exponent mappings

$$n_{\text{Barrow}} = \Delta - 2, \quad n_{\text{Tsallis}} = 2\delta_T - 4. \quad (2)$$

Both reduce to standard HDE [1] at $n = -2$ (Barrow $\Delta = 0$, Tsallis $\delta_T = 1$). The two “different” entropies are therefore the same one-parameter family, seen from opposite sides of the standard-HDE point: Barrow explores $n > -2$ (flatter $\rho(L)$), Tsallis $\delta_T < 1$ explores $n < -2$ (steeper $\rho(L)$).

We stress that this equivalence is asserted at the level of the resulting dark-energy *density scaling* $\rho_{\text{DE}} \propto L^n$, which is what the background expansion and all our observables depend on—not necessarily at the level of the horizon first law. Barrow and Tsallis entropies are distinct functionals, and a full thermodynamic derivation from each (through the apparent-horizon first law $dE = TdS$) need not yield identical modified Friedmann equations term by term. What we use, and all that the fits require, is that with a future-event-horizon cutoff both produce a power-law density with the exponent mappings above; two models with the same n share the same background dynamics and are observationally identical here regardless of the entropy functional they descend from. The unification is thus phenomenological and exact at the level of the observable, and we make no claim of equivalence at the level of the underlying field theory.

A single background equation covers both. Writing $x = \ln a$ and $\Omega \equiv \Omega_{\text{DE}}$,

$$\frac{d\Omega}{dx} = \Omega(1 - \Omega) \left[1 - \frac{n}{c} \Omega^{-1/n} \right], \quad (3)$$

with c the standard HDE constant. Equation (3) reproduces the original Barrow equation of motion to machine precision, and at $n = -2$, $c = 1$, $\Omega_{\text{DE}0} = 0.69$ yields the exact standard-HDE value $w_0 = -0.891$. The corresponding equation of state follows analytically from Eq. (3) and continuity,

$$w(\Omega) = \frac{-1 + (n/c) \Omega^{-1/n}}{3}, \quad (4)$$

with the correct limits $w \rightarrow -1/3$ as $\Omega \rightarrow 0$ (matter era) and $w \rightarrow -1$ as $\Omega \rightarrow 1$ (de Sitter future).

A numerical remark, relevant to reproducibility: the future event horizon makes de Sitter a *forward* attractor, so integrating Eq. (3) forward from today is unstable (errors grow toward $\Omega \rightarrow 1$). We integrate instead by shooting from a modest future point $x \simeq 3$ and solving both directions, tuning $\Omega(x_{\text{start}})$ to hit $\Omega_{\text{DE}0}$; and we obtain w from the analytic form Eq. (4) rather than by numerical differentiation.

2.2 Rényi as a genuine generalization

Rényi holographic dark energy does not reduce to a power law. With

$$\rho_{\text{DE}} = \frac{3C^2}{8\pi L^2 (1 + \delta\pi L^2)}, \quad (5)$$

the cutoff L does not drop out of the background dynamics, and one is left with a genuinely coupled two-dimensional system in (Ω_{DE}, L) . We solve it by backward integration from the de Sitter future attractor, shooting on the normalization to hit $\Omega_{\text{DE}0}$; in the $\delta \rightarrow 0$ limit this converges to standard HDE ($w_0 = -0.889$ at $\delta = 0.01$). Here C^2 plays the role of the squared HDE constant, $C^2 = c^2$, so that $C^2 = 1$ recovers $c = 1$ and $w_0 \simeq -0.891$; we adopt this normalization to match the power-law family.

Although Rényi is not a power law, its perturbations reduce to an *effective local exponent*, a fact we exploit in Sec. 4: writing $\rho_{\text{DE}} = f(L)$,

$$n_{\text{eff}}(L) \equiv \frac{L f'(L)}{f(L)} = -2 - \frac{2\delta\pi L^2}{1 + \delta\pi L^2}, \quad (6)$$

which runs monotonically from -2 (small L , standard HDE) to -4 (large L , Rényi-dominated) and is bounded in $[-4, -2]$.

2.3 Kaniadakis as a fourth member

Kaniadakis holographic dark energy, built from the relativistic κ -deformed entropy [8], is a further one-parameter extension. Like Rényi it is not a power law: the κ -deformation of the area law modifies the Friedmann equation and yields a coupled background whose equation of state can be quintessence-like, phantom-like, or cross the phantom divide, with the acceleration transition at $z \simeq 0.6$. It has been shown to be a competitive but not preferred alternative to Λ CDM against DESI DR2, cosmic chronometers, and supernovae [8, 9]. We include it as a fourth member of the family; being a distinct coupled background it is treated like Rényi rather than through the power-law ODE Eq. (3). Crucially for Sec. 4, Kaniadakis holographic dark energy uses the future event horizon as its infrared cutoff, so—whatever its background behavior—its perturbations fall under the same closed-channel theorem: the entropy deformation changes the density law, not the null-geodesic definition of the horizon.

3 The data discriminate at the background level

3.1 Data and likelihood

We fit each model to real DESI DR2 anisotropic BAO (six tracers over $z = 0.51\text{--}2.33$, D_M/r_d and D_H/r_d with their correlations) [7] and Planck 2018 CMB distance priors [5, 6], the compressed likelihood in $\{R, \ell_A, \omega_b\}$ with the Chen et al. inverse covariance. The shift parameter, acoustic scale, and sound horizon are

$$R = \sqrt{\Omega_m} \int_0^{z_*} \frac{dz}{E(z)}, \quad \ell_A = \pi \frac{D_M(z_*)}{r_s(z_*)}, \quad (7)$$

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{E(z)} dz, \quad c_s = \frac{1}{\sqrt{3(1+R_b)}}, \quad (8)$$

with $R_b = 3\omega_b/(4\omega_\gamma)/(1+z)$ and z_* from the Hu–Sugiyama fitting formula. We compute r_s using the analytic matter-plus-radiation $E(z)$ above z_* , where dark energy is dynamically negligible; this is essential, as retaining the residual dark-energy term makes the high-redshift r_s integral converge incorrectly. The likelihood is validated in Ref. [13]: pure- Λ CDM Planck parameters reproduce $R = 1.750$ and $\ell_A = 300.4$ (to 0.3%).

We profile over (Ω_m, H_0, c) and, for the final results, over ω_b as well (Sec. 3.5).

3.2 Rényi at the CMB

Extending Rényi HDE to $z_* \simeq 1090$ requires care. The future-event-horizon variable $L \rightarrow 0$ at high redshift, so the Rényi density Eq. (5) grows in absolute terms ($\rho_{\text{DE}} \sim C^2/L^2 \sim (1+z)^2$). This is not a pathology: since matter scales as $(1+z)^3$ and radiation as $(1+z)^4$, the fractional contribution $\Omega_{\text{DE}}(z) \rightarrow 0$ monotonically ($\Omega_{\text{DE}} \sim 2 \times 10^{-4}$ at z_*), so the model has a small, integrable early dark-energy tail rather than a divergence in any observable. What the growing ρ_{DE} does cause is amplified numerical noise if the horizon system is integrated naively through the near-singular regime. We handle it with a matching redshift z_{match} : below it we evolve the full coupled Rényi system; above it we use the analytic matter-plus-radiation $E(z)$ retaining the $(1+z)^2$ dark-energy tail rather than setting it to zero, so no physical early-DE contribution to $r_s(z_*)$ is neglected. The split is therefore a numerical device, not a physical transition. The result is insensitive to z_{match} once the tail is carried; dropping it instead (an earlier, cruder choice) leaves a z_{match} -dependent residual of ~ 0.2 in ℓ_A that converges away for $z_{\text{match}} \gtrsim 50$. The procedure reproduces $z_* = 1091.8$ and, in the $\delta \rightarrow 0$ limit with $C^2 = 1$, recovers standard HDE ($w_0 = -0.887$, $R = 1.70$, $\ell_A = 300.1$). The full numerical mechanics are given in the Appendix.

3.3 Discrimination result

Table 1 collects the best-fit total χ^2 (DESI BAO + CMB distance priors, H_0 free) across the family. Three state-

Table 1: Best-fit total χ^2 (DESI DR2 BAO + Planck CMB distance priors, H_0 free) across the horizon-entropy family. **Numbers in red are provisional**, carried over from an earlier fixed- ω_b run, and are to be regenerated with the ω_b -marginalized pipeline (Sec. 3.5) before submission. Qualitative ordering is expected to be robust; absolute values are not yet final.

Model	deformation	total χ^2
Λ CDM	—	~ 10
Tsallis	$\delta_T = 0.64$	24.5
Tsallis	$\delta_T = 0.70$	27.3
Standard HDE	$n = -2$	38.1
Barrow	$\Delta = 0.1$	43.8
Rényi	$\delta = 0.1$	BAO ~ 11; CMB via split competitive; ΛCDM favored
Kaniadakis	K small	

ments stand out, none of which is available from single-model studies.

(i) *Standard HDE is already disfavored, before any deformation.* With a future-event-horizon cutoff, standard HDE ($n = -2$) sits in strong tension with DESI DR2 BAO geometry: $\chi^2_{\text{BAO}} \simeq 47$ versus $\simeq 10$ for Λ CDM, a $\Delta\chi^2 \simeq 37$ penalty independent of the CMB. This is consistent with the post-DESI finding that standard HDE is strained.

(ii) *The direction of deformation is decisive.* Steepening $\rho(L)$ (Tsallis $\delta_T < 1$, $n < -2$) relieves the tension, primarily by repairing the CMB fit; flattening it (Barrow $\Delta > 0$, $n > -2$) makes matters worse. The best HDE variant is Tsallis $\delta_T \simeq 0.64$, with total $\chi^2 \simeq 24.5$ (from **29** $\rightarrow$ **2.6** in the CMB), while Barrow $\Delta = 0.1$ rises to $\simeq 43.8$.

(iii) *None beats Λ CDM.* The best HDE fit remains $\Delta\chi^2 \simeq 14$ above Λ CDM. Generalized entropy is thus not monolithic: the data select *steeper* $\rho(L)$ but do not favor any member of the family over the cosmological constant.

3.4 The H_0 preference

Freeing H_0 improves every HDE fit substantially, toward the local distance-ladder value $H_0 \simeq 74$: standard HDE moves from $\chi^2 \simeq 75.5$ at $H_0 = 67.4$ to $\simeq 38.1$ at $H_0 \simeq 74$. This is a known property of future-event-horizon HDE and is Hubble-tension-adjacent. We flag one honest caveat: our CMB enters only as distance priors, which are more permissive in H_0 than the full Planck likelihood, so the strength of the $H_0 \simeq 74$ preference is partly a property of the compressed likelihood. The *relative* model ordering is robust to this.

3.5 Status of the numbers

The χ^2 values in Table 1 are provisional. They derive from a run in which ω_b was fixed at 0.02237. Because

$\sigma(\ell_A)$ is small, the CMB likelihood is strongly sensitive to ω_b through r_s : a model preferring ω_b shifted by as little as $\sim 4 \times 10^{-4}$ —well within the Planck uncertainty—incur a spurious penalty of tens in χ^2 when ω_b is held fixed. This penalty is *not* uniform across models, since different deformations reshape the low-redshift distance and therefore pull ω_b in different directions. The final table must therefore profile (marginalize) over ω_b per model. We expect the qualitative ordering (steeper helps, flatter hurts, none beats Λ CDM) to survive, but the absolute values, and in principle the ordering, must be confirmed with the ω_b -marginalized pipeline. The ω_b profiling itself is a one-dimensional minimization inside each model’s χ^2 , validated on self-consistent mocks; what remains is to run it on the fully calibrated distance-prior pipeline of Ref. [13].

4 The structure-growth channel is closed

Recent analyses call for confronting horizon-entropy dark energy at the perturbative, structure-growth level, on the expectation that $f\sigma_8$ adds information beyond the background. We show analytically that for event-horizon HDE it does not: the models are, to the precision that matters, smooth, and the growth rate is degenerate with $H(z)$.

4.1 Growth depends on the cosmology only through $H(a)$

For smooth (non-clustering) dark energy the linear growth factor obeys, in e -folds $N = \ln a$,

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \Omega_m(a) D = 0. \quad (9)$$

Every cosmology-dependent term is a function of $H(a)$ alone: both the friction $2 + H'/H$ and the source $\frac{3}{2}\Omega_m(a) = \frac{3}{2}\Omega_{m0}a^{-3}/(H/H_0)^2$. There is no independent dark-energy degree of freedom. Two models with identical $H(a)$ produce identical $D(a)$, hence identical $f\sigma_8(z)$. Numerically, a CPL model tuned to mimic Λ CDM to within 0.5% in $H(z)$ over $0 < z < 2$ differs by at most 0.5% in $f\sigma_8$ —the same order as the background difference, far below current ~ 5 –10% growth-rate errors. The growth observable inherits the background (non-)difference and adds nothing.

4.2 The sound speed cannot be adiabatic

Independent information would require dark energy to cluster, i.e. a rest-frame sound speed $c_s^2 < 1$. But the adiabatic value $c_a^2 = w - \dot{w}/[3H(1+w)]$ is negative or singular for $w < -1/3$ —generic for dark energy, and singular near the phantom line several variants approach—so it must be discarded on stability grounds (imposing it drives sub-horizon δ_{DE} to blow up). HDE has no Lagrangian kinetic

term from which to read a rest-frame c_s^2 ; the density is defined non-locally by a horizon. The effective sound speed is thus whatever the horizon prescription implies, which we now compute.

4.3 Horizon perturbations phase-average away

Physically, because the future event horizon is defined by an integral along the future light cone, its value at the observer sums the gravitational potential over a long null path that threads many alternating crests and troughs of a small-scale perturbation; these contributions largely cancel, so the horizon—and the dark-energy density tied to it—barely responds to sub-horizon structure. We now make this quantitative.

With $\rho_{DE} \propto R_h^n$, exactly $\delta_{DE} = n(\delta R_h/R_h)$. The future event horizon is a null construct; perturbing the radial null geodesic in Newtonian gauge, the comoving horizon acquires

$$\delta\chi_k(t) = 2e^{ikx} \int_0^{\chi_h} \Phi_k(u) e^{iku} du, \quad u = \chi(t, t'), \quad (10)$$

the comoving distance along the future ray. Here Φ_k is the Newtonian-gauge potential, which coincides with the Bardeen invariant Φ_A ; since the null ray is itself a geometric (gauge-independent) object, $\delta R_h/R_h$ is built entirely from gauge-invariant ingredients and is not an artifact of the conformal Newtonian slicing. A residual gauge shift in the dark-energy density contrast is $\mathcal{O}(aH/k)(1+w)v_{DE}$; with v_{DE} sourced at $\mathcal{O}(\Phi)$ this is of order δ_m/k^3 , the same order as the result itself, so no slicing can promote δ_{DE} to a non-decaying mode—there is none to transform into. For a sub-horizon mode ($k \gg aH$) the kernel e^{iku} oscillates across the integration range; integration by parts with any smooth envelope gives $\int_0^{\chi_h} \Phi_k e^{iku} du = \mathcal{O}(1/k)$. Hence $|\delta_{DE}| \sim (aH/k)|\Phi|$: the potential is rapidly *phase-averaged* along the future light cone, and a small-scale fluctuation contributes essentially nothing to the non-local horizon integral.

Relative to the matter that actually clusters, the suppression is stronger still. Sub-horizon, Φ is sourced by matter through Poisson, $k^2\Phi = -4\pi G a^2 \rho_m \delta_m$, so $\Phi \sim \delta_m/k^2$ and

$$\frac{\delta_{DE}}{\delta_m} \sim \frac{1}{k^3} \quad (k \gg aH). \quad (11)$$

Dark energy is smooth to third order in aH/k relative to matter. We record this as the central result of this section:

No-clustering theorem for event-horizon dark energy. For any horizon-entropy dark-energy model with a future-event-horizon cutoff, the sub-horizon dark-energy density contrast is suppressed as $\delta_{DE}/\delta_m \sim (aH/k)^3$; the linear growth of structure is governed by $H(a)$ alone, and $f\sigma_8$

carries no information about the model beyond its background expansion.

(We state the $1/k$ scaling of δ_{DE} directly rather than dressing it as an effective $c_s^2 = 1$; a genuine $c_s^2 = 1$ fluid gives a $1/k^2$ forced amplitude, so the horizon suppression relative to Φ is in fact weaker than $c_s^2 = 1$, while relative to matter it is $1/k^3$.)

The integration by parts deserves its boundary term made explicit, since it is local and not phase-averaged. Writing $\int_0^{\chi_h} \Phi_k e^{iku} du = [\Phi_k e^{iku}/ik]_0^{\chi_h} - (1/ik) \int_0^{\chi_h} \Phi'_k e^{iku} du$, the surface piece at the observer, $-\Phi_k(0)/ik$, does not oscillate in k and so is not averaged away. But it is itself $\mathcal{O}(1/k)$: $\Phi_k(0)$ is the finite local potential, not a quantity that grows with k . The far endpoint $\Phi_k(\chi_h) e^{ik\chi_h}/ik$ oscillates in k and averages down further. Both the surface and bulk contributions are therefore $\mathcal{O}(1/k)$, so $\delta R_h \sim \mathcal{O}(1/k)$ and the scaling of Eq. (11) is unchanged. A degradation to $1/k^2$ would require $\delta R_h = \mathcal{O}(1)$, i.e. $\Phi_k(0) = \mathcal{O}(k)$, which is unphysical. A numerical evaluation of Eq. (10) with a nonzero local potential $\Phi_k(0) \neq 0$ confirms $|\delta\chi_k| \propto k^{-1}$ exactly, the surface term supplying the leading $1/k$ rather than any $1/k^0$ piece.

We keep the leading light-ray phase term; local metric corrections at the observation point are also $\mathcal{O}(aH/k)$ and subleading. A three-closure numerical integration across $k = 1\text{--}1000$ confirms the picture: an imposed $c_s^2 = 1$ leaves sub-horizon δ_{DE} at $\sim 10^{-9}$ (smooth), the adiabatic $c_s^2 < 0$ blows up (unphysical), and the horizon closure tracks the smooth branch.

4.4 Extension to Rényi

Rényi modifies the density law $f(L)$, not the definition of L : the cutoff remains the future event horizon, a geometric null construct. Its perturbation is $\delta_{\text{DE}} = n_{\text{eff}}(L) (\delta L/L)$ with n_{eff} the bounded quantity of Eq. (6). The scale dependence lives entirely in δL —the same oscillatory horizon integral, which is kinematic (it arises from null propagation, $ds^2 = 0 \Rightarrow dx = dt/a$) and untouched by the coupled background. The coupled system alters only bounded, smooth, non-oscillatory quantities (the horizon size, the potential envelope, and the n_{eff} prefactor). Integration by parts still gives $\mathcal{O}(1/k)$, so $|\delta_{\text{DE}}| = |n_{\text{eff}}| \mathcal{O}(aH/k) \rightarrow 0$ sub-horizon. Rényi is smooth on the same footing as the power-law family.

4.5 Relation to recent sound-speed constraints

It might appear that our conclusion is in tension with recent work that, for the first time, constrains the clustering of dynamical dark energy from current data. Combining DESI DR2 BAO with Planck 2018 and the Union3 supernovae, Yang et al. [10] report a bound on the effective sound speed, $\log_{10} c_s^2 = -3.00^{+2.9}_{-0.99}$, in a $w_0 w_a$ CDM background within the parameterized-post-Friedmann and

effective-field-theory descriptions. Taken at face value this says the data can now “see” dark-energy clustering; how can the growth channel be closed?

There is no contradiction, and the distinction is the substance of our result. That analysis treats c_s^2 as a *free* parameter, made estimable by the time-varying equation of state, which breaks the degeneracy between $(1+w)$ and c_s^2 . Its constraint is a statement about what the data could reveal *if* dark energy has an adjustable sound speed. Horizon-entropy dark energy is not of that kind. Its density is fixed non-locally by a horizon, with no independent stress sector to carry a free c_s^2 ; the effective sound speed is not a parameter to be fit but is *determined* by the horizon prescription, and for an event-horizon cut-off that determination is the smooth branch, Eq. (11). In short: the sound-speed constraint says *if it clusters, we can measure it*; our result says *event-horizon holographic dark energy does not cluster, so there is nothing to measure*. The two are complementary halves of the same physical statement, and together they sharpen it—the growth data that can now probe a free sound speed have no purchase on a model whose sound speed the horizon has already fixed.

4.6 A consistency check: the growth index

The smooth limit makes a concrete, falsifiable prediction for the growth index γ , defined by $f = \Omega_m^\gamma$. Mehrabi et al. [11] derive the asymptotic index for holographic dark energy in two distinct regimes. For *homogeneous* (non-clustering) holographic dark energy the general result $\gamma_\infty = 3(w_\infty - 1)/(6w_\infty - 5)$, evaluated at the holographic asymptotic equation of state $w_\infty \rightarrow -1/3$, gives

$$\gamma_\infty^{\text{homog}} \simeq \frac{4}{7}, \quad (12)$$

about 5% above the Λ CDM value 6/11. For *fully clustered* holographic dark energy a different source term applies and the index becomes $\gamma_\infty^{\text{clust}} \simeq 3(1 - c_{\text{eff}}^2)/7$, which at $c_{\text{eff}}^2 = 0$ gives 3/7. These are two separate branches, not endpoints of one interpolation.

Our result places event-horizon holographic dark energy squarely in the homogeneous branch: the $1/k^3$ suppression means the dark-energy density contrast never sources the matter growth equation ($\delta_{\text{DE}} \simeq 0$), so growth is governed by $H(a)$ and $\Omega_m(a)$ alone. The asymptotic homogeneous value $\gamma_\infty^{\text{homog}} \simeq 4/7$ is set by the high- z holographic equation of state $w \rightarrow -1/3$; it is *not* directly the observable index, because in these models the late-time attractor is de Sitter ($w \rightarrow -1$), not $w = -1/3$. To obtain the observable prediction we integrate the growth equation numerically along the event-horizon background ($w = -\frac{1}{3}(1 + 2\sqrt{\Omega_{\text{DE}}})$ for the $c = 1$ cutoff), reading off $\gamma(z) = \ln f / \ln \Omega_m$:

z	2.0	1.0	0.5	0.3	0.0
$\gamma(z)$	0.562	0.560	0.560	0.560	0.561

The observable index is remarkably flat at $\gamma \simeq 0.56$ across the range where growth is measured—close to but distinct from the asymptotic $4/7 \simeq 0.571$, and sitting about 0.015 above the Λ CDM value $6/11 \simeq 0.545$. The prediction is thus model-external and falsifiable: the entire event-horizon class predicts $\gamma \simeq 0.56$, weakly z -dependent and insensitive to the entropy choice (which affects only $H(a)$, already constrained at the background level). A measurement pinning γ to the Λ CDM value 0.545 or below would disfavour the class; a value near the fully clustered $3/7 \simeq 0.43$ would indicate the perturbations do source growth, contradicting the closed channel. Separating 0.56 from 0.545 requires the precision of a Euclid-era growth survey, but the boundary is clean.

4.7 The one condition

The derivation assumes the infrared cutoff is the *future event horizon*. If instead a variant uses the Hubble horizon $L = 1/H$ (or a mixed cutoff), then $\delta L = -\delta H/H^2$ is a *local* quantity with no oscillatory kernel and no $1/k$ suppression, and dark energy could genuinely cluster. The precise statement is therefore: *event-horizon horizon-entropy dark energy is smooth on sub-horizon scales, and $f\sigma_8$ carries no information beyond $H(z)$, if and only if the model uses a light-ray (event-horizon) cutoff*. Our models use the future event horizon; the result holds. (Hubble-cutoff HDE independently fails to produce late-time acceleration in the presence of dark matter, so the event horizon is the standard choice.) A Hubble- or mixed-cutoff construction is the one place in this analysis where the perturbation channel could reopen—a well-defined open problem.

It is worth noting that the event-horizon cutoff on which our result depends is also the one the data prefer. Combining expansion and growth-rate data across the three most common cutoffs, Akhlaghi et al. [12] find the Ricci and Granda–Oliveros prescriptions disfavored—they predict a non-negligible early-time dark-energy density—while the future-event-horizon cutoff remains consistent with the combined data. Our scope is therefore not a convenience: the cutoff for which the growth channel is provably closed is the same cutoff the observations single out.

5 Discussion

We have shown two things about horizon-entropy dark energy that a single-model fit cannot reveal. At the background level, a uniform confrontation with real DESI DR2 BAO and Planck distance priors *discriminates*: steeper $\rho(L)$ (Tsallis $\delta_T < 1$) relieves the standard-HDE tension, flatter $\rho(L)$ (Barrow $\Delta > 0$) worsens it, and none of the family is preferred over Λ CDM. At the perturbation level, the structure-growth channel is *closed*: for any event-horizon member, sub-horizon dark-energy perturbations are suppressed as $1/k^3$ relative to matter by

phase-averaging along the future light cone, so $f\sigma_8$ is degenerate with the background $H(z)$ and adds no discriminating power.

The two results are complementary. The first says the models are testable now, at the background level, and shows what the test yields. The second says the verdict cannot be overturned by adding growth data: one cannot rescue a background-disfavored member by appealing to a distinctive clustering signature, because there is none. For this class the proposed structure-growth confrontation is structurally empty—a useful thing to know before investing in it.

Two limitations bound these conclusions. The background numbers of Sec. 3 are provisional pending the ω_b -marginalized run (Sec. 3.5); we expect the qualitative ordering to survive but state plainly that it must be confirmed. And the closed-channel result is conditional on an event-horizon cutoff (Sec. 4.7); the Hubble-cutoff case is left open and is the natural next target, being the one construction in which growth data could become informative.

Two matters of scope should be stated explicitly. First, we treat these models as *late-time* modifications of the dark-energy sector: the entropy deformation governs the infrared, horizon-scale dark energy, while early-universe thermodynamics—recombination, the sound horizon $r_s(z_*)$, and the thermal history—are taken to be standard. This is what licenses the use of the Planck distance priors (R, l_A, ω_b) , which are extracted assuming standard pre-recombination physics. It is a genuine restriction: a construction in which the modified entropy also deforms the apparent-horizon thermodynamics of the radiation era would shift $r_s(z_*)$ and invalidate the compressed likelihood, and such a model would require a full Boltzmann treatment rather than distance priors. Our results apply to the phenomenologically standard case where the deformation is confined to the dark sector at low redshift; we do not claim them for early-modified variants. Second, the phantom members (Tsallis $\delta_T < 1$, $n < -2$) can cross $w = -1$; where they approach a future Big Rip at finite t_{rip} , the event-horizon integral $L = a \int_t^{t_{\text{rip}}} dt'/a$ terminates at t_{rip} rather than at infinity, and the background shooting must impose the finite-time boundary there. This is well-defined but changes the future boundary condition; we have restricted the fitted range to members with a de Sitter future, and flag the phantom/Big-Rip branch as requiring separate boundary treatment.

Acknowledgments

This work uses public DESI DR2 BAO and Planck data products and builds on the detectability framework and validated distance-prior likelihood of Ref. [13].

A Numerical mechanics of the horizon integration

We collect the implementation details here to keep the main text focused on results. Two points matter for reproducibility.

Background shooting. The future event horizon makes de Sitter a forward attractor, so integrating Eq. (3) forward from today is numerically unstable: errors grow toward $\Omega \rightarrow 1$. We instead shoot from a modest future point $x = \ln a \simeq 3$, integrating in both directions and tuning $\Omega(x_{\text{start}})$ to hit the measured Ω_{DE0} at $x = 0$. The equation of state is taken from the analytic form Eq. (4) rather than by numerically differentiating $\Omega(x)$, which avoids amplifying integration noise.

Rényi high-redshift matching. For the Rényi (and Kaniadakis) coupled backgrounds the density $\rho_{\text{DE}} \sim C^2/L^2 \sim (1+z)^2$ grows toward high redshift while $\Omega_{\text{DE}} \rightarrow 0$. Rather than integrate the stiff coupled system through this regime, we match at z_{match} : the full coupled system below, and the analytic matter-plus-radiation $E(z)$ above, *retaining* the $(1+z)^2$ dark-energy tail so no physical early-DE contribution to $r_s(z_*)$ is dropped. The sound-horizon integral uses matter and radiation only for the acoustic sound speed, $c_s = [3(1+R_b)]^{-1/2}$ with $R_b = 3\omega_b/4\omega_\gamma/(1+z)$, and z_* from the Hu-Sugiyama fitting formula. With the tail retained the compressed observables are stable against z_{match} ; we use $z_{\text{match}} \gtrsim 50$ and verify convergence.

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